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BELL INEQUALITIES – 17.12.2025

NICO MACHAT, DANIEL KÖSTLER, MICHAEL SCHLEINZER

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1 Abstract

The aim of this experiment was to demonstrate that quantum mechanics cannot be a local realistic theory. This can be shown by violating a so-called Bell inequality. In our case, we use the CHSH inequality, which is a specific form of Bell inequalities.

Einstein, Podolsky and Rosen published a paper in which they argued that quantum mechanics must be either incomplete or not a local realistic theory. Several decades later, Bell showed that if one assumes quantum mechanics to be incomplete, this will require the existence of hidden variables. He further demonstrated that for any local realistic theory, a Bell inequality must be satisfied.

In the case of the CHSH inequality, this means that the introduced parameter S has to fulfill the condition $S \leq 2$. If this inequality is violated, local hidden variable theories are ruled out. Consequently, quantum mechanics cannot be described as a local realistic theory.

This is demonstrated in the present experiment by generating entangled photon pairs via spontaneous parametric down-conversion (SPDC). The photons are detected in two separate arms, and the corresponding expectation values of the joint measurements are determined. From these expectation values, the CHSH parameter S is calculated and used to test the Bell inequality.

Furthermore, the correlation curves of individual measurement bases are recorded in this experiment. From these correlation curves, it is also possible to infer whether the prepared states are entangled or separable.

2 Introduction

2.1 Preparation of the Required Quantum States

In this experiment, three different quantum states are prepared: the entangled Bell states $|\phi_+\rangle$ and $|\phi_-\rangle$, as well as the separable product state $|HH\rangle$. The pump laser emits horizontally polarized light. By inserting a half-wave plate into the pump beam, the polarization of the pump light can be adjusted in order to prepare the different states.

For the preparation of the $|\phi_+\rangle$ state, the half-wave plate is oriented such that the pump light is linearly polarized at 45° , resulting in equal horizontal and vertical polarization components with the same phase.

By rotating the half-wave plate by 180° , the pump light remains linearly polarized at 45° , but a relative phase shift of π between the horizontal and vertical components is introduced. This leads to the preparation of the $|\phi_-\rangle$ state.

If the half-wave plate is removed entirely, the pump light remains purely horizontally polarized. In this case, only one of the two nonlinear crystals is pumped, resulting in the separable product state $|HH\rangle$.

The differently polarized pump light then impinges on the nonlinear crystals, where the corresponding two-photon states are generated via spontaneous parametric down-conversion.

The three states are:

$$|\phi_+\rangle = \frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle)$$

$$|\phi_-\rangle = \frac{1}{\sqrt{2}}(|HH\rangle - |VV\rangle)$$

$$|HH\rangle$$

2.2 Theoretical Background

2.2.1 Initial Situation

Our starting point is the following. Quantum mechanics predicts the phenomenon of entanglement, which implies that two entangled particles exhibit very strong correlations. In particular, when one particle is measured, the state of the other particle is perfectly correlated with the outcome of the first measurement. This prediction follows directly from quantum mechanics and can be verified experimentally.

According to the theory, these correlations persist regardless of the spatial separation between the particles. Even if the particles are separated by a large distance, the measurement of one particle immediately determines the state of the other. Quantum mechanics itself imposes no condition on the distance between the particles for this correlation to hold.

However, this leads to a conceptual problem: if the particles are spatially separated and no interaction or communication between them is allowed, it is unclear how one particle can “know” that the other has been measured. This apparent instantaneous correlation challenges the principle of locality, suggesting the presence of a non-local influence.

In order to avoid abandoning locality, one may alternatively assume the existence of hidden variables. In this picture, the measurement outcomes are already predetermined when the particles are created but are governed by variables that are not accessible within standard quantum mechanics. This idea was proposed in the seminal paper by Einstein, Podolsky, and Rosen.

Thus, the phenomenon of entanglement cannot be explained without either giving up local realism—allowing for non-local correlations—or assuming that quantum mechanics is incomplete and must be supplemented by hidden variables. This dilemma constitutes the starting point of the Bell argument.

In his seminal work, Bell challenged the concept of local hidden-variable theories by demonstrating their logical consequences. He derived an inequality that must be satisfied by any theory based on local hidden variables. If this inequality is violated, such hidden-variable theories cannot exist.

Subsequently, Clauser, Horne, Shimony, and Holt formulated a modified version of Bell's inequality, known as the CHSH inequality, which is experimentally accessible. They also demonstrated experimentally that this inequality is violated by quantum-mechanical predictions. In the following, we derive the CHSH inequality used in our experiment.

Important to note is that Bell's theorem does not rule out hidden variables per se and therefore does not prove that quantum mechanics is a complete theory. What Bell's theorem demonstrates is that no theory based on local hidden variables can reproduce the experimentally observed correlations. Consequently, any viable theory must abandon locality. However, this does not imply that the underlying influence must be strictly instantaneous. Bell inequalities only impose a lower bound on the speed at which such correlations would have to be established, which exceeds the speed of light, but does not require infinite or instantaneous propagation.

2.2.2 Derivation of the CHSH Bell Inequality

The experiment consists of two spatially separated measurement stations, conventionally referred to as Alice and Bob. Each station is equipped with two single-photon detectors, preceded by a polarization analyzer. As described in Section 2.1, the photon pairs are prepared in an entangled quantum state. One photon of each pair is directed to Alice, while the other photon is sent to Bob.

At each station, the photon first encounters a polarizer, which transmits light only for a specific polarization direction. Since a photon is an indivisible quantum object, it cannot be partially transmitted and partially blocked. Instead, it is either fully transmitted or fully absorbed by the polarizer. Depending on the orientation of the polarizer relative to the photon's polarization state, the photon is transmitted with a certain probability or blocked with the complementary probability.

Alice and Bob each have two possible settings for their polarization analyzers. Alice can choose between the angles α and α' , while Bob can choose between β and β' . We assume that Alice's measurement outcome depends only on her local setting and on a hidden variable λ . The measurement result is dichotomic and can take only the values $+1$ or -1 , where $+1$ corresponds to the photon being transmitted through the polarizer and -1 corresponds to the photon being blocked and thus not detected.

Accordingly, Alice's outcomes are described by the functions $A(\alpha, \lambda)$ and $A(\alpha', \lambda)$, each taking values in $\{+1, -1\}$. Analogously, Bob's measurement outcomes are described by the functions $B(\beta, \lambda)$ and $B(\beta', \lambda)$, which also take values in $\{+1, -1\}$ and depend only on Bob's local setting and the hidden variable λ .

The hidden variable λ is distributed according to a normalized probability distribution $\rho(\lambda)$ such that:

$$\rho(\lambda) \geq 0 \quad \text{and} \quad \int \rho(\lambda) d\lambda = 1$$

From this, the expectation value for a given combination of the measurement settings α and β can be defined:

$$E(\alpha, \beta) = \int d\lambda \rho(\lambda) A(\alpha, \lambda) B(\beta, \lambda)$$

From this, we can define the CHSH parameter, which is given by the following linear combination of expectation values:

$$S = E(\alpha, \beta) - E(\alpha, \beta') + E(\alpha', \beta) + E(\alpha', \beta')$$

Inserting the expectation values into the definition of the CHSH parameter yields:

$$\begin{aligned} S &= \int d\lambda \rho(\lambda) [A(\alpha, \lambda)B(\beta, \lambda) - A(\alpha, \lambda)B(\beta', \lambda) + A(\alpha', \lambda)B(\beta, \lambda) + A(\alpha', \lambda)B(\beta', \lambda)] \\ &= \int d\lambda \rho(\lambda) S(\lambda) \end{aligned}$$

$$\begin{aligned} S(\lambda) &= A(\alpha, \lambda)B(\beta, \lambda) - A(\alpha, \lambda)B(\beta', \lambda) + A(\alpha', \lambda)B(\beta, \lambda) + A(\alpha', \lambda)B(\beta', \lambda) \\ &= A_\alpha B_\beta + A_{\alpha'} B_\beta - A_\alpha B_{\beta'} + A_{\alpha'} B_{\beta'} \end{aligned}$$

For brevity, the notation has been simplified by writing $A_\alpha = A(\alpha, \lambda)$ and similarly for the other terms. All quantities remain functions of the corresponding measurement settings and the hidden variable λ .

In the following, the terms A_α and $A_{\alpha'}$ can be factored out, allowing the expression to be rewritten in a form that makes the dependence on the hidden variable λ explicit.

$$S(\lambda) = A_\alpha (B_\beta - B_{\beta'}) + A_{\alpha'} (B_\beta + B_{\beta'})$$

For a given value of the hidden variable λ , the measurement outcomes corresponding to the settings β and β' can only take the values $+1$ or -1 . Consequently, only two distinct cases have to be considered:

$$\text{Fall 1: } B_\beta = B_{\beta'} \Rightarrow S(\lambda) = \pm 2 A_{\alpha'} = \pm 2$$

$$\text{Fall 2: } B_\beta = -B_{\beta'} \Rightarrow S(\lambda) = \pm 2 A_\alpha = \pm 2$$

In the following, we take the absolute value of the parameter S . By applying the triangle inequality, an upper bound for S is obtained:

$$S = \int d\lambda \rho(\lambda) S(\lambda)$$

$$|S| = \left| \int d\lambda \rho(\lambda) S(\lambda) \right|$$

$$|S| \leq \int d\lambda \rho(\lambda) |S(\lambda)|$$

$$|S(\lambda)| = 2$$

$$|S| \leq 2 \int d\lambda \rho(\lambda) = 2$$

$$|S| \leq 2$$

The CHSH inequality $|S| \leq 2$ follows directly from the assumptions of locality and realism. Realism requires that all four measurement outcomes exist simultaneously for a given hidden variable λ , while locality demands that each outcome depends only on the local measurement setting. Under these assumptions, the CHSH parameter is strictly bounded by 2. An experimental violation of this bound therefore implies that local realistic descriptions of nature are incompatible with quantum mechanical predictions.

2.2.3 Evaluation of the Expectation Values

To determine the expectation values, the coincidence counts corresponding to the possible measurement outcomes are required. A coincidence count corresponds to the case in which both photons are transmitted through their respective polarizers and detected, yielding a correlated outcome. However, events in which both photons are absorbed are not directly observable, and single detections cannot be unambiguously assigned to correlated photon pairs due to background and single-photon events.

To reconstruct all required contributions, four separate measurements are performed for each pair of analyzer settings. First, both polarizers are set to the angles α and β . Second, both analyzers are rotated by 90° , corresponding to the orthogonal settings. Third, only Bob's analyzer is rotated by 90° , and finally only Alice's analyzer is rotated by 90° . These four measurements allow all joint outcomes to be inferred and combined to calculate the expectation value for the given angle combination.

Each measurement is performed over the same integration time of 10 seconds to ensure consistent normalization. The expectation value is then obtained by combining the corresponding counts with the appropriate signs and normalizing by the total number of events.

The expectation value for a given combination of analyzer angles is therefore calculated as follows, where C denotes the measured (corrected) coincidence count rate for the corresponding angle combination.

$$E(\alpha, \beta) = \frac{C(\alpha, \beta) - C(\alpha, \beta_\perp) - C(\alpha_\perp, \beta) + C(\alpha_\perp, \beta_\perp)}{C(\alpha, \beta) + C(\alpha, \beta_\perp) + C(\alpha_\perp, \beta) + C(\alpha_\perp, \beta_\perp)}$$

From this, it is immediately evident that a total of 16 measurements are required. The evaluation of the CHSH inequality involves four expectation values. Since each detector (or polarizer) can be set to two different angles, four possible angle combinations arise, corresponding to the four expectation values. For each expectation value, four separate measurements are necessary in order to determine all required coincidence counts. Consequently, the experiment requires 16 measurements with 16 different combinations of analyzer angles.

2.2.4 Statistical uncertainty due to Poisson statistics

At the level of individual detection events, the measurement outcome is binary: either a photon is detected or it is not, and either a coincidence between both detectors occurs within the coincidence window or it does not. When measuring over a finite time interval (here 10 seconds), the number of detected coincidences is therefore not identical in repeated measurements, but fluctuates due to the intrinsically random nature of the photon detection process. These fluctuations are described by Poisson statistics.

If the same measurement were repeated many times under identical conditions, one would obtain a distribution of coincidence counts rather than a single fixed value. There is thus no unique “true” count number; instead, the measurement outcome is characterized by a mean value and a statistical spread. For large count numbers, the relative statistical uncertainty decreases, which is why long integration times and high count rates are advantageous. Nevertheless, a finite statistical uncertainty always remains.

In order to properly assess the measured value of the CHSH parameter, it is therefore necessary to quantify this statistical uncertainty. The width of the distribution is given by the standard deviation, which determines the range within which repeated measurements are expected to fluctuate. Approximately 68 % of all measurement outcomes lie within one standard deviation around the mean value. By comparing the measured value e.g. $S = 2.574$ to the classical bound $S = 2$ in units of this standard deviation, one can determine how unlikely it is that the observed violation arises purely from statistical fluctuations. This provides a quantitative measure of the significance of the Bell inequality violation.

The standard deviation and the number of standard deviations by which the measured result deviates from the classical bound can be calculated using the following formula:

$$\Delta S(\alpha, \alpha', \beta, \beta') = \sqrt{\sum_{a=\alpha, \alpha'} \sum_{b=\beta, \beta'} [\Delta E(a, b)]^2}$$

$$\Delta E(a, b) = 2 \frac{[C(a, b) + C(a_{\perp}, b_{\perp})][C(a, b_{\perp}) + C(a_{\perp}, b)]}{[C(a, b) + C(a, b_{\perp}) + C(a_{\perp}, b) + C(a_{\perp}, b_{\perp})]^2} \sqrt{\frac{1}{C(a, b) + C(a_{\perp}, b_{\perp})} + \frac{1}{C(a, b_{\perp}) + C(a_{\perp}, b)}}$$

$$n_{\Delta} = \frac{S - 2}{\Delta S}$$

3 Experimental setup

The experiment was performed using a quED entanglement demonstrator. Polarization-entangled photon pairs were generated via spontaneous parametric down-conversion (SPDC) in two type-I BBO crystals, which were pumped by a blue laser diode. The optical axes of the two crystals are oriented orthogonally, allowing the generation of polarization-entangled photon pairs when the pump polarization is set to 45° .

The down-converted photons are emitted into two spatially separated modes and collected by single-mode optical fibers. In each arm, a polarizer is placed to analyze the polarization of the photons. Long-pass filters are used to suppress residual pump light. The photons are detected by avalanche photodiodes, and single as well as coincidence count rates are recorded by the control electronics.

By setting the polarizers to different angular configurations, polarization correlations between the two photons are measured and used to evaluate the CHSH Bell inequality.

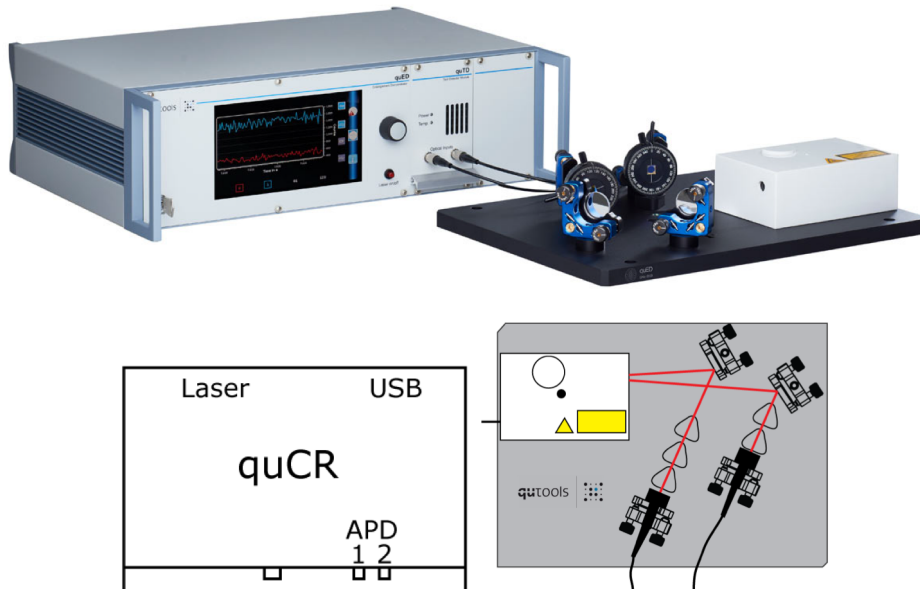


Figure 1 Experimental Setup

4 Violation of the CHSH Bell Inequality

4.1 Measurement Procedure

In this experiment, we demonstrate the violation of a Bell inequality. This is achieved by using three different quantum states, as described in Section 2.1. The experimental setup is prepared accordingly in order to generate the respective states. The measurements are then performed automatically by the experimental apparatus.

As derived in Section 2.2, the evaluation of the CHSH Bell inequality requires four expectation values. Each expectation value is obtained from four different measurement settings. Consequently, a total of 16 different measurements, corresponding to 16 different combinations of polarization angles, are required.

For each angle combination, the number of coincidence counts is recorded over an integration time of 10 seconds. From these coincidence counts, the expectation values are calculated automatically by the control software, which subsequently determines the CHSH parameter S .

The statistical uncertainty is evaluated assuming Poisson statistics for the photon counting process. Based on this uncertainty, the deviation of the measured S -value from the classical bound $S = 2$ is calculated and expressed in terms of the number of standard deviations. This quantifies the significance of the observed violation of the Bell inequality.

The tables corresponding to the individual measurements list the polarization angle settings of the analyzers, the total single count rates of the two detectors, the measured coincidence counts, and the corrected coincidence counts. The coincidence counts correspond to the number of detection events registered by both detectors within a fixed coincidence time $\Delta t = 30 \text{ ns}$. In most cases, such events originate from a common photon pair. However, due to the high overall count rates, it is statistically unavoidable that uncorrelated photons occasionally fall within the coincidence time window. Therefore, the expected number of accidental coincidences is estimated automatically by the apparatus based on the single count rates, the measured coincidence counts, and the assumption of Poissonian counting statistics. This estimated background is subtracted from the measured coincidences, yielding the corrected coincidence counts used for the further evaluation.

4.2 Optimization of Coincidence Rates and Polarization Balance

Before performing the actual measurements, the polarization analyzers were adjusted in order to optimize the coincidence count rate. This procedure ensures an optimal overlap between the polarization correlations of the generated photon pairs and the measurement bases of the detectors. In addition to maximizing the coincidence rate, the alignment was chosen such that the coincidence rates for the relevant polarization components were comparable, indicating a balanced detection efficiency for the different polarization states (e.g. horizontal–horizontal and vertical–vertical components). This optimization step is essential to ensure reliable and unbiased coincidence measurements and to reduce systematic effects in the subsequent evaluation of the Bell inequality.

The alignment was carried out for both entangled states, $|\phi_+ \rangle$ - state and $|\phi_- \rangle$ - state. For the separable $|HH \rangle$ - state, no further optimization was necessary.

4.3 Qualitative Expectation of Coincidence Rates

This short section provides a qualitative estimation of the expected coincidence rates for different prepared quantum states and measurement bases. The three states considered in the experiment were summarized in the table, and the corresponding measurement bases were chosen such that both detectors measured in the same basis, in mutually orthogonal bases, and in the $\pm 45^\circ$ basis. In the latter case, measurements were considered both for identical analyzer orientations and for orientations shifted by 90° . This comparison allows for an intuitive understanding of how the coincidence rates depend on the prepared quantum state and the chosen measurement basis.

Measurement basis	$ \phi_+ \rangle$	$ \phi_- \rangle$	$ HH \rangle$
$ HH \rangle, VV \rangle$	High	High	High / Low
$ HV \rangle, VH \rangle$	Low	Low	Low
$ ++ \rangle, -- \rangle$	High	Low	Middle
$ +- \rangle, -+ \rangle$	Low	High	Middle

Table 1 Expected coincidence behavior for different quantum states in various measurement bases.

- **Identical measurement bases (HH or VV):**

When both analyzers are set to the same polarization (HH or VV), the coincidence rate is high for both since both states contain the HH and VV components with equal probability. The coincidence rate for the HH-state is high for HH measurements and low (approximately zero) for VV measurements because VV is not possible.

- **Orthogonal measurement bases (HV or VH):**

When the analyzers are set to orthogonal polarizations, the coincidence rate is low (close to zero) for both entangled states, since only HH and VV components are present. The same holds for the HH product state, for which no coincidences are expected in orthogonal bases.

- **Measurement in the $++$ or $--$ basis:**

In this measurement basis, both detectors probe whether the photons are in the same superposition state with equal amplitude and phase. For the $|\phi_+ \rangle$ state, this condition is fulfilled, leading to a high coincidence rate, whereas for the $|\phi_- \rangle$ state the relative phase difference results in destructive interference and therefore a low coincidence rate.

- **Measurement in the $+-$ or $-+$ basis:**

In this case, the detectors probe opposite superposition states. Consequently, the coincidence rate is low for the $|\phi_+ \rangle$ state and high for the $|\phi_- \rangle$ state, reflecting the opposite phase relation between the two components of the entangled state.

The following table shows the measured coincidence rates for the different measurement bases. These data were recorded prior to the final optimization of the coincidence rates, i.e. before the polarization analyzers were adjusted to maximize the coincidence counts. As a result, some values may appear slightly irregular. Nevertheless, the overall trend of the coincidence rates is clearly visible and agrees well with the qualitative expectations summarized in the previous table.

Measurement basis	$ \phi_+ \rangle$	$ \phi_- \rangle$	$ HH \rangle$
$ HH \rangle, VV \rangle$	3107, 1232	1599, 1404	2568, 15
$ HV \rangle, HV \rangle$	64, 32	44, 31	34, 5
$ ++ \rangle, -- \rangle$	1852, 1719	71, 59	885, 639
$ +- \rangle, -+ \rangle$	331, 395	1525, 1363	540, 603

Table 2 Measured coincidence counts for the different measurement bases prior to optimization.

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4.4 Measurement with the $|\phi_+ \rangle$ - state

4.4.1 Automated Calculation of the CHSH Parameter

This measurement was performed using the $|\phi_+ \rangle$ - state. The integration time for each polarization angle combination was 10 seconds.

X (deg)	Y (deg)	Rate Det. 1	Rate Det. 2	Coincidences	Corrected Coincidences
0.0	22.5	513 634	442 082	20 191	19 736
0.0	67.5	515 104	402 334	4 740	4 325
0.0	112.5	515 539	400 738	3 465	3 051
0.0	157.5	516 473	444 409	19 263	18 803
45.0	22.5	407 902	440 386	18 589	18 229
45.0	67.5	408 559	402 495	20 000	19 671
45.0	112.5	408 279	400 681	5 475	5 147
45.0	157.5	408 599	443 678	4 543	4 180
90.0	22.5	284 704	441 192	3 188	2 936
90.0	67.5	283 855	401 803	19 819	19 590
90.0	112.5	284 500	400 625	21 405	21 177
90.0	157.5	284 528	443 289	4 642	4 389
135.0	22.5	390 404	440 355	5 976	5 632
135.0	67.5	390 591	402 792	4 475	4 160
135.0	112.5	391 981	401 600	18 842	18 527
135.0	157.5	391 206	442 156	20 423	20 077

Table 3 Measurement angles and coincidence rates for the $|\phi_+ \rangle$ - state

The values of the CHSH parameter S , its associated uncertainty, and the number of standard deviations calculated by the apparatus are listed in the following table.

CHSH Result for S	2.574
CHSH Error for S	0.007
Number of standard deviations	82

Table 4 Results of the CHSH measurement for the $|\phi_+ \rangle$ - state

4.4.2 Manual Calculation of the CHSH Parameter

As described in Section 2.2, the S-value can be calculated using the following expression:

$$S(\alpha, \alpha', \beta, \beta') = E(\alpha, \beta) + E(\alpha', \beta) - E(\alpha, \beta') + E(\alpha', \beta')$$

$$E(\alpha, \beta) = \frac{C(\alpha, \beta) - C(\alpha, \beta_{\perp}) - C(\alpha_{\perp}, \beta) + C(\alpha_{\perp}, \beta_{\perp})}{C(\alpha, \beta) + C(\alpha, \beta_{\perp}) + C(\alpha_{\perp}, \beta) + C(\alpha_{\perp}, \beta_{\perp})}$$

For the calculation of the expectation values, the corrected coincidence counts listed in Table 3 are used.

$$\begin{aligned} E(0^\circ, 22.5^\circ) &= \frac{C(0^\circ, 22.5^\circ) - C(0^\circ, 112.5^\circ) - C(90^\circ, 22.5^\circ) + C(90^\circ, 112.5^\circ)}{C(0^\circ, 22.5^\circ) + C(0^\circ, 112.5^\circ) + C(90^\circ, 22.5^\circ) + C(90^\circ, 112.5^\circ)} \\ &= \frac{19736 - 3051 - 2936 + 21177}{19736 + 3051 + 2936 + 21177} = \frac{34926}{46900} \approx \mathbf{0.745} \end{aligned}$$

$$\begin{aligned} E(45^\circ, 22.5^\circ) &= \frac{C(45^\circ, 22.5^\circ) - C(45^\circ, 112.5^\circ) - C(135^\circ, 22.5^\circ) + C(135^\circ, 112.5^\circ)}{C(45^\circ, 22.5^\circ) + C(45^\circ, 112.5^\circ) + C(135^\circ, 22.5^\circ) + C(135^\circ, 112.5^\circ)} \\ &= \frac{18229 - 5147 - 5632 + 18527}{18229 + 5147 + 5632 + 18527} = \frac{25977}{47535} \approx \mathbf{0.547} \end{aligned}$$

$$\begin{aligned} E(0^\circ, 67.5^\circ) &= \frac{C(0^\circ, 67.5^\circ) - C(0^\circ, 157.5^\circ) - C(90^\circ, 67.5^\circ) + C(90^\circ, 157.5^\circ)}{C(0^\circ, 67.5^\circ) + C(0^\circ, 157.5^\circ) + C(90^\circ, 67.5^\circ) + C(90^\circ, 157.5^\circ)} \\ &= \frac{4325 - 18803 - 19590 + 4389}{4325 + 18803 + 19590 + 4389} = \frac{-29679}{47107} \approx \mathbf{-0.630} \end{aligned}$$

$$\begin{aligned} E(45^\circ, 67.5^\circ) &= \frac{C(45^\circ, 67.5^\circ) - C(45^\circ, 157.5^\circ) - C(135^\circ, 67.5^\circ) + C(135^\circ, 157.5^\circ)}{C(45^\circ, 67.5^\circ) + C(45^\circ, 157.5^\circ) + C(135^\circ, 67.5^\circ) + C(135^\circ, 157.5^\circ)} \\ &= \frac{19671 - 4180 - 4160 + 20077}{19671 + 4180 + 4160 + 20077} = \frac{31408}{48088} \approx \mathbf{0.653} \end{aligned}$$

$$\begin{aligned} S(0^\circ, 45^\circ, 22.5^\circ, 67.5^\circ) &= E(0^\circ, 22.5^\circ) + E(45^\circ, 22.5^\circ) - E(0^\circ, 67.5^\circ) + E(45^\circ, 67.5^\circ) \\ &= 0.745 + 0.547 - (-0.630) + 0.653 \approx \mathbf{2.574} \geq 2 \end{aligned}$$

As described in Section 2.4, the measured coincidence counts follow Poisson statistics. This leads to a statistical uncertainty that must be evaluated in order to assess the reliability of the result. As outlined in Section 2.4, this uncertainty is calculated using the following two expressions:

$$\Delta S(\alpha, \alpha', \beta, \beta') = \sqrt{\sum_{a=\alpha, \alpha'} \sum_{b=\beta, \beta'} [\Delta E(a, b)]^2}$$

$$\Delta E(a, b) = 2 \frac{[C(a, b) + C(a_{\perp}, b_{\perp})][C(a, b_{\perp}) + C(a_{\perp}, b)]}{[C(a, b) + C(a, b_{\perp}) + C(a_{\perp}, b) + C(a_{\perp}, b_{\perp})]^2} \sqrt{\frac{1}{C(a, b) + C(a_{\perp}, b_{\perp})} + \frac{1}{C(a, b_{\perp}) + C(a_{\perp}, b)}}$$

For the calculation of the uncertainties, the corrected coincidence counts listed in Table 3 are used.

$$\Delta E(0^\circ, 22.5^\circ) = \frac{2 (19736 + 21177)(3051 + 2936)}{(19736 + 3051 + 2936 + 21177)^2} \sqrt{\frac{1}{19736 + 21177} + \frac{1}{3051 + 2936}} \approx \mathbf{0.00308}$$

$$\Delta E(45^\circ, 22.5^\circ) = \frac{2 (18229 + 18527)(5147 + 5632)}{(18229 + 5147 + 5632 + 18527)^2} \sqrt{\frac{1}{18229 + 18527} + \frac{1}{5147 + 5632}} \approx \mathbf{0.00384}$$

$$\Delta E(0^\circ, 67.5^\circ) = \frac{2 (4325 + 4389)(18803 + 19590)}{(4325 + 18803 + 19590 + 4389)^2} \sqrt{\frac{1}{4325 + 4389} + \frac{1}{18803 + 19590}} \approx \mathbf{0.00358}$$

$$\Delta E(45^\circ, 67.5^\circ) = \frac{2 (19671 + 20077)(4180 + 4160)}{(19671 + 4180 + 4160 + 20077)^2} \sqrt{\frac{1}{19671 + 20077} + \frac{1}{4180 + 4160}} \approx \mathbf{0.00345}$$

$$\Delta S(0^\circ, 45^\circ, 22.5^\circ, 67.5^\circ) = \sqrt{(0.00308)^2 + (0.00384)^2 + (0.00358)^2 + (0.00345)^2} \approx \mathbf{0.007}$$

$$n_{\Delta} = \frac{S - 2}{\Delta S} = \frac{2.574 - 2}{0.007} \approx \mathbf{82}$$

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4.5 Measurement with the $|\phi_- \rangle$ - state

Note: For the $|\phi_- \rangle$ - state and the $|HH \rangle$ - state, only the automated calculation performed by the experimental apparatus is used, and no manual calculation is carried out.

This measurement was performed using the $|\phi_- \rangle$ - state. The integration time for each polarization angle combination was 10 seconds.

X (deg)	Y (deg)	Rate Det. 1	Rate Det. 2	Coincidences	Corrected Coincidences
0.0	22.5	389 853	469 881	21 585	21 218
0.0	67.5	391 206	429 537	5 597	5 260
0.0	112.5	389 969	425 879	2 830	2 497
0.0	157.5	390 681	469 576	18 579	18 212
45.0	22.5	359 470	466 882	6 009	5 673
45.0	67.5	359 252	429 711	2 868	2 559
45.0	112.5	359 404	425 237	18 478	18 172
45.0	157.5	358 910	466 926	21 363	21 027
90.0	22.5	323 385	467 646	2 523	2 220
90.0	67.5	323 877	429 120	18 514	18 236
90.0	112.5	322 760	426 165	21 333	21 057
90.0	157.5	322 807	466 668	5 519	5 217
135.0	22.5	357 230	467 944	18 988	18 653
135.0	67.5	356 611	429 670	21 622	21 315
135.0	112.5	357 900	426 539	5 529	5 223
135.0	157.5	355 966	466 566	3 247	2 914

Table 5 Measurement angles and coincidence rates for the $|\phi_- \rangle$ - state

The values of the CHSH parameter S , its associated uncertainty, and the number of standard deviations calculated by the apparatus are listed in the following table.

CHSH Result for S	2.667
CHSH Error for S	0.007
Number of standard deviations	98

Table 6 Results of the CHSH measurement for the $|\phi_- \rangle$ - state

4.6 Measurement with the $|HH\rangle$ - state

This measurement was performed using the $|HH\rangle$ - state. The integration time for each polarization angle combination was 10 seconds.

X (deg)	Y (deg)	Rate Det. 1	Rate Det. 2	Coincidences	Corrected Coincidences
0.0	22.5	661 904	936 762	28 779	27 538
0.0	67.5	660 752	242 832	7 074	6 753
0.0	112.5	660 660	143 028	4 043	3 854
0.0	157.5	662 323	871 250	26 665	25 510
45.0	22.5	360 205	938 678	17 068	16 391
45.0	67.5	360 794	244 869	5 046	4 869
45.0	112.5	361 333	144 218	1 487	1 382
45.0	157.5	361 364	875 233	14 163	13 530
90.0	22.5	9 655	936 174	145	126
90.0	67.5	9 583	243 355	240	235
90.0	112.5	9 626	143 625	152	149
90.0	157.5	9 535	873 170	84	67
135.0	22.5	339 427	938 151	14 575	13 938
135.0	67.5	341 725	244 674	2 842	2 674
135.0	112.5	341 133	143 969	3 050	2 951
135.0	157.5	341 467	872 234	15 219	14 623

Table 7 Measurement angles and coincidence rates for the $|HH\rangle$ - state

The values of the CHSH parameter S , its associated uncertainty, and the number of standard deviations calculated by the apparatus are listed in the following table.

CHSH Result for S	1.538
CHSH Error for S	0.01
Number of standard deviations	-48

Table 8 Results of the CHSH measurement for the $|HH\rangle$ - state

4.7 Interpretation

$|\phi_+ \rangle$ - state:

The values for the CHSH parameter calculated by the apparatus are in exact agreement with our manual calculations. The results clearly demonstrate a violation of the Bell inequality, with a measured value of $S = 2.574$, corresponding to an excess of $S - 2 = 0.574$ above the classical bound. The statistical uncertainty, derived from Poissonian counting statistics, is $\Delta S = 0.007$. Consequently, the measured value exceeds the classical limit by approximately 82 standard deviations. This indicates that the observed violation cannot be attributed to statistical fluctuations and can be excluded with extremely high confidence.

$|\phi_- \rangle$ - state:

The same conclusion applies to the $|\phi_- \rangle$ - state. In this case, an even stronger violation of the Bell inequality is observed, with a measured value of $S = 2.667$ and the same statistical uncertainty as for the $|\phi_+ \rangle$ - state, namely $\Delta S = 0.007$. This corresponds to a deviation of approximately 98 standard deviations from the classical bound. Consequently, it can be concluded with very high confidence that this result is not caused by statistical fluctuations.

$|HH \rangle$ - state:

For the $|HH \rangle$ - state, an expected result is obtained as well. Since this state is not entangled but represents a simple product state in which both photons are always horizontally polarized, no violation of the Bell inequality is observed. The measured CHSH parameter is $S = 1.538$ with an uncertainty of $\Delta S = 0.01$, corresponding to a deviation of 48 standard deviations from the classical bound $S = 2$. This confirms that the $|HH \rangle$ - state exhibits purely classical correlations, as expected.

5 Measurement of Polarization Correlation Curves

In the second part of the experiment, correlation curves were recorded for all three prepared quantum states. After each state was generated and properly aligned, the correlation measurements were performed directly by the apparatus. The analyzer angle was varied in steps of 1° , and for each angular position the coincidence counts were recorded over an integration time of 2 seconds. This procedure was repeated for the full angular range, resulting in the correlation curves evaluated automatically by the measurement system.

No changes were made to the experimental setup. As described in the previous experiment, the individual quantum states were prepared in the same manner, and the setup was left unchanged. Only the correlation curves were recorded using the measurement apparatus.

5.1 Determination of the Visibility

From the measured correlation curves it is possible to determine whether a quantum state is entangled or separable by evaluating the visibility. For this purpose, the visibility is calculated for all measured correlation curves corresponding to the different analyzer settings.

If the state is entangled, high-contrast correlation fringes are observed consistently for all relevant measurement settings, resulting in visibilities exceeding the threshold value of 0.707.

In contrast, for a separable (product) state, at least one of the correlation curves exhibits a significantly reduced or vanishing modulation, leading to a visibility below 0.707.

Therefore, if at least one of the measured correlation curves has a visibility smaller than 0.707, the state is identified as non-entangled. Only if all correlation curves exhibit visibilities above this threshold can the state be classified as entangled.

The visibility is calculated according to the following formula:

$$V = \frac{C_{\max} - C_{\min}}{C_{\max} + C_{\min}}$$

5.2 Correlation curves measured with the quCR software

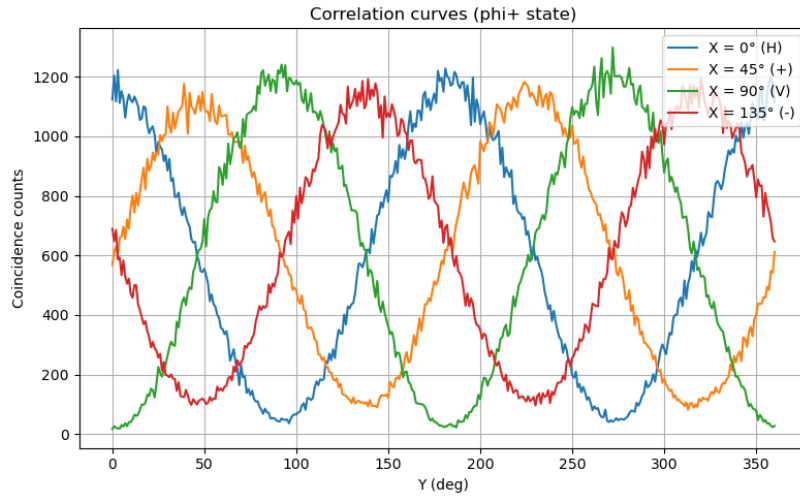


Figure 2 Correlation curves for the $|\phi_+ \rangle -$ state

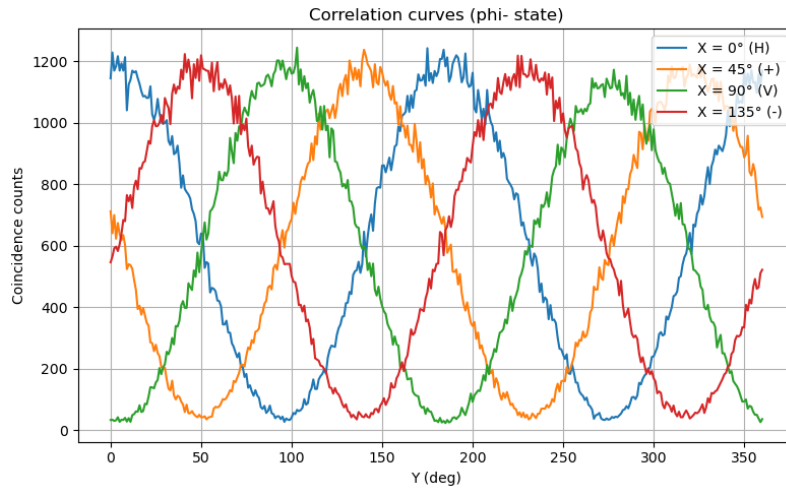


Figure 3 Correlation curves for the $|\phi_- \rangle -$ state

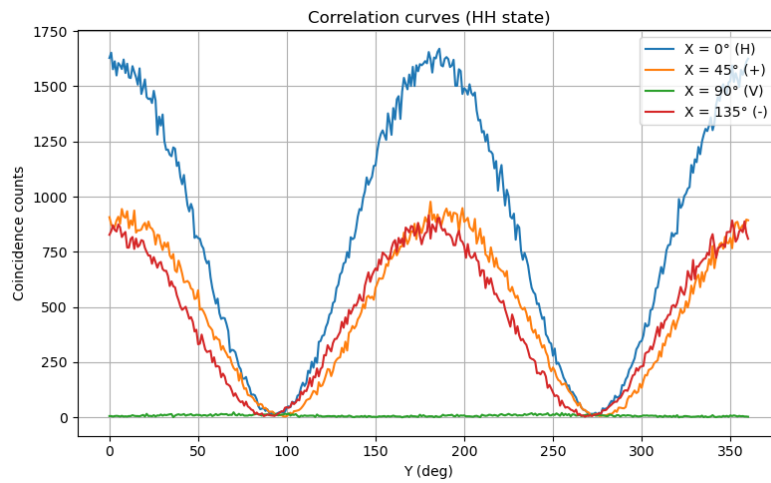


Figure 4 Correlation curves for the $|\text{HH} \rangle -$ state

5.3 Interpretation

$|\phi_+ \rangle$ - state:

$$V_H = V_V = \frac{1200 - 40}{1200 + 40} = 0.935 \quad V_+ = V_- = \frac{1200 - 40}{1200 + 40} = 0.935$$

For the $|\phi_+ \rangle$ - state, the measured correlation curve shows a clear sinusoidal dependence on the analyzer angle with high visibility. This behavior is characteristic of a maximally entangled state and reflects the strong quantum correlations between the two photons. The positions of maxima and minima are in good agreement with the predictions of quantum mechanics, indicating constructive and destructive correlations depending on the relative analyzer settings.

$|\phi_- \rangle$ - state:

$$V_H = V_V = \frac{1200 - 40}{1200 + 40} = 0.935 \quad V_+ = V_- = \frac{1200 - 40}{1200 + 40} = 0.935$$

The correlation curve for the $|\phi_- \rangle$ - state exhibits a sinusoidal modulation similar to that of the $|\phi_+ \rangle$ - state. The visibility is high for all measured correlation curves and clearly exceeds the threshold value of 0.707. This confirms that the $|\phi_- \rangle$ - state is also maximally entangled and displays strong non-classical correlations.

$|HH \rangle$ - state:

$$V_H = \frac{1625 - 40}{1625 + 40} \quad V_V = 0 \quad V_+ = V_- = \frac{1200 - 40}{1200 + 40} = 1$$

In contrast, the correlation curve obtained for the $|HH \rangle$ - state shows a fundamentally different behavior. Since the $|HH \rangle$ - state is a separable product state with both photons always horizontally polarized, no entanglement is present. Consequently, the coincidence rates follow a classical angular dependence and do not exhibit the pronounced sinusoidal modulation observed for the entangled states. In addition, the visibility for the VV basis measurement is found to be zero. This confirms the absence of non-classical correlations and explains why the Bell inequality is not violated for this state.

6 Appendix

List of Equipment

qutools quED Set which includes:

- Optics part like mirrors and fibers
- Electronics part with the quCR control unit
- Motors for the polarisator

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